do what they want, and take progress one step at a time. This he could do because of the immense talent he found in his men, the individual integrity of the members and the cohesion of the team. Hierarchy was there for basic administrative reasons; but Sarabhai and Murthy did not allow the typical Indian bureaucracy to interfere with work.

This I realised at my first day at work. I walked in and Kalam asked me, 'What are you interested in doing?'

I said I will do anything he wants me to, provided it was engineering. Kalam suggested I create some devices on pyrotechnics, the science of explosives.

'Why don't you develop an explosive bolt?' he said.

I knew little about that, but the sound of the term made it clear to me that it was a bolt that would explode. I did some reading and found that an explosive bolt is primarily used by test pilots to snap their seat belts in case of an emergency ejection. A small explosion goes off inside a bolt that holds the seat belt and, poof, he is ejected. Our application of the device would be for an adaptation called cable cutter. This chute has to be deployed at high altitudes after a payload is taken into the upper atmosphere by a sounding rocket. The payload has to be recovered without damage, and hence the parachute.

I started drawing the plan for a 0.5-inch diameter bolt. The idea was to create a cavity in which there would be an explosive charge and a squib. When given an electric charge, the squib—which is a narrow 'U' shaped thread-like structure with a small quantity of an explosive—sparks and triggers a slightly bigger explosion in the bolt, snapping it. Two weeks later, I was making calculations for the thickness of the cavity and the kind of charge to be used when a shadow fell on the floor a few metres ahead of my desk.

Seated as I was with the drawing board resting on the chair, the paraphernalia virtually having nailed me to the seat, I looked up and saw a handsome man in a white kurta-pajama walking in. The man had grace and radiated energy. Behind him, Kalam and a couple of others had emerged,